

Reconstructing and Re-Evaluating BSFR-SH: A Blockchain-Enabled Ransomware Defense Framework

Course Project Report

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Abstract—BSFR-SH claims a blockchain-protected, honeypot-driven detection and recovery framework for ransomware in smart healthcare, reporting 98.98% accuracy and 0.990 F1 on the BitcoinHeist dataset. We built a complete, from-scratch implementation of all five of its algorithms, both of its private blockchains under real pBFT consensus, and the honeypot feature pipeline the paper specifies but never defines. Reproducing the paper’s own evaluation protocol, our best model reaches 0.9442 accuracy / 0.9697 F1 – 4.56 points below the published figure, and an ablation over both candidate leakage sources we identified narrows but does not close a 3.58-point residual gap. We further identify four independent defects spanning the paper’s evaluation methodology, security argument, and preprocessing specification. The framework’s central idea – chain-protecting a detector’s ground truth and a system’s backups from the same adversary – is sound; several of the paper’s supporting claims measurably are not.

I. Introduction

BSFR-SH [1] proposes a blockchain-enabled defense against ransomware for smart healthcare networks. Its contribution is architectural: two independent, permissioned blockchains, one holding encrypted data backups and one holding ransomware signatures and behavioural features harvested from a honeypot, both committed through pBFT consensus among a fixed set of cloud servers. The pitch is that an attacker who compromises a protected system cannot also tamper with the detector’s ground truth or destroy the means of recovery, because both are chain-certified and the chain is Byzantine-fault-tolerant against a minority of colluding servers. On top of this architecture the paper reports a ransomware detector that reaches 98.98% accuracy and 0.990 F1 on the BitcoinHeist dataset, beating four prior published detectors, and a consensus/backup pipeline that commits 1500 transactions in under seven seconds.

No artifact accompanies the paper: no code, no declared hyperparameters, no train/test protocol beyond “90/10,” no definition of what a honeypot-derived “signature” or “feature” actually is. Reproducing a paper under those conditions means first building the thing it describes closely enough that its own claims become checkable, and only then checking them. This report does both. We built the full system – crypto layer, two blockchains, pBFT consensus, honeypot sample synthesis and feature extraction, a dual-backend ML detection module, a mitigation state machine, and a recovery path – against

the paper’s own algorithm listings, and we ran its own evaluation protocol wherever the paper specifies one. Where it does not – which, as §III details, is often – we designed the missing piece deliberately, documented why, and kept the design decision separate from any claim about reproducing the paper’s numbers: a gap the paper leaves open is not evidence against the paper, and we treat the two categories, undefined design and measurable defect, differently throughout.

We state the result up front rather than at the end. We found five independent defects in the paper, spanning its evaluation methodology, its security argument, its preprocessing specification, its reported throughput claim, and the correspondence between what its framework is designed to detect and what it is actually evaluated on. And its headline detection result does not reproduce: under the paper’s own protocol, our best model falls 4.56 accuracy points short of the published figure, and a leakage ablation designed to find an innocent explanation narrows that gap to 3.58 points without closing it. Neither finding is presented as a failure of this project – both are measured, bounded, and reported with baselines. The paper’s central architectural idea survives scrutiny better than its supporting claims do, and §V says so explicitly.

II. Paper Summary

BSFR-SH defines five algorithms across two independent blockchains, BC_{DTBU} (data backups) and BC_{SigRW} (ransomware signatures and features), each committed by a four-node pBFT [2] cluster ($n = 4$, tolerating $f = 1$ Byzantine replica, commit threshold $2f + 1 = 3$).

Algorithm 1 (backup creation). A protected system $SY S_i$ ships its backup DT_{BU} to a cloud server CS_i over an established session key. CS_i encrypts it into transactions $E_{KU}(Tx_m)$, assembles a block, and the pBFT leader drives consensus to append it to BC_{DTBU} .

Algorithm 2 (data collection). A honeypot HP_{RW} is deployed and harvested; raw data DT_{RW} is cleaned to DT_{RWC} , from which a signature Sig_{RW} and a feature vector FT_{RW} are “generated” (the paper’s own word, and its only specification of either); the result is encrypted, blocked, and committed to BC_{SigRW} through the same pipeline shape as Algorithm 1.

Algorithm 3 (detection). A detection module DM_{CS_i} decrypts BC_{SigRW} , trains four classifiers – random forest, logistic regression, decision tree, k -nearest-neighbours – builds a normal profile $NProf$ and an abnormal profile $AProf$, and runs continuous detection.

Algorithm 4 (mitigation). On a positive detection, SYS_i is isolated and remediated by one of three cases: Case-1 erases the ransomware in place; Case-2 formats the system and restores it via Algorithm 5; Case-3, if the demanded ransom is less than the assessed value of the data, pays it and requests the decryption key.

Algorithm 5 (recovery). SYS_i ’s backups are located in BC_{DTBU} , decrypted by the cloud server holding the wrapping key, relayed through a second server, and delivered back to SYS_i .

Section V argues five security properties informally: mutual-authentication session keys defeat replay/MITM/impersonation; deleted post-registration credentials defeat privileged-insider and stolen-verifier attacks; pBFT resists 51% attacks, selfish mining, and (in a permissioned deployment) Sybil attacks; chain immutability resists DoS, manipulation, and leakage; and the two-chain split isolates detection from recovery.

Section VII evaluates the ML component on the Bitcoin-Heist Ransomware Address Dataset [3] (2,916,697 rows, 1.42% ransomware), resampled to a 90% ransomware / 10% benign split, reporting accuracy and F1 for all four classifiers and comparing the best against four prior published detectors. It separately benchmarks consensus and block construction on both chains at three cases (5/10/15 blocks $\times$ 100 transactions) on a single Windows machine (Intel i5 9th gen, 8 GB RAM) running four miner-node processes in Java, reporting wall-clock time and transactions-per-second for each case and chain.

Neither evaluation states what a reproduction would need to check it independently. Section VII gives no hyperparameters for any of the four classifiers, no cross-validation scheme, and no random seed; “90/10 split” is the entire description of the sampling protocol. The timing benchmark states block and transaction counts but not a single transaction’s payload size, so every second in Fig. 6 is, on the page, a function of an unstated constant. Both gaps are filled explicitly in our implementation (declared hyperparameters, declared payload size, a swept sensitivity range) rather than guessed once and left implicit, so that our own reported numbers are reproducible from our own configuration even where the paper’s are not.

III. Implementation

The paper leaves several parts of its own design unspecified. Our rule throughout was to fill each gap deliberately and document the rationale rather than silently “fixing” the paper, so that every departure from the published design is a recorded decision, not an unremarked substitution. This section covers six of those design decisions, chosen because each one is load-bearing for whether the

framework, once built, actually does what the paper says it does.

The scope of what had to be built to reach that point is worth stating once, here, before the individual decisions: a full crypto layer (ECDSA, AEAD, hybrid key wrapping, a Merkle tree with its second-preimage weakness closed, a mutual-authentication session protocol), two independent blockchain implementations under a real four-node pBFT cluster with view-change on leader timeout, a honeypot sample synthesizer and feature extractor, a dual-backend detection module, a mitigation state machine, and a recovery path – exercised by an adversarial test suite that includes tampered blocks, wrong signatures, colluding and equivocating Byzantine replicas, replay and man-in-the-middle attempts against the session protocol, and Sybil identity swarms against the permissioned membership. Every module in this section is one piece of that whole; the whole is what makes the honeypot pipeline’s end-to-end result in §VI possible at all.

A. Hybrid encryption for transaction payloads

Algorithm 1 line 2 and Algorithm 2 line 7 write transaction payloads as $E_{KU_{CS_i}}(Tx)$ – encrypted directly under the receiving cloud server’s public key. Elliptic-curve public-key encryption is not designed to carry bulk payloads, and healthcare backups are not small. We implement standard hybrid encryption instead: AES-256-GCM over the payload, with the symmetric data key wrapped by ECIES under KU_{CS_i} . The semantics are unchanged – only CS_i can decrypt – but the substitution introduces a seam the original single-primitive scheme does not have. A wrapped key and the payload it protects are now two separate byte strings inside one transaction, and if they are merely adjacent rather than bound, an attacker who forges nothing can lift the wrapped key from one transaction and attach it to another’s payload: every block signature still verifies, because the signature covers whatever bytes the block happens to contain, while CS_i ends up decrypting data the honest sender never submitted. That defeats the paper’s own §V-4 data-manipulation claim without breaking a single cryptographic primitive. We close it by binding the wrap’s authenticated data to the transaction’s metadata, and the payload’s authenticated data to that metadata plus the wrapped-key bytes, so separating either half breaks a GCM authentication tag.

B. Mutual authentication and session establishment

Section IV-A and §V-1 defer session establishment to “any standard mutual authentication and key establishment mechanism.” We designed one: ECDH over secp256r1, ECDSA-signed transcripts with RFC 6979 deterministic nonces [8] (the same primitive the paper already mandates for signatures), timestamps, and message nonces. Implementing it surfaced two defects in our own first draft, either of which would have made §V-1’s claims false rather than merely unspecified. First, the

initiator’s signature did not name the intended responder, so a captured session opening addressed to one cloud server replayed verbatim as a genuine-looking request to a different one – impersonation requiring no key material at all. Second, a timestamp acceptance window bounds only how long a captured message stays acceptable; it does not prevent replay within that window. We fixed the first by signing the responder’s identity into the initiator’s message, and the second with a seen-nonce cache keyed by (peer, nonce) held for twice the timestamp window. Both properties are now pinned by dedicated replay and man-in-the-middle tests.

§V is entirely informal prose – five paragraphs of argument, no ROR model, no BAN logic, no tool-checked proof – for a paper whose author’s other work almost always includes one. We verified this protocol, which is our own design and not the paper’s, with Scyther [9] under its default Dolev-Yao adversary. Table I maps §V-1’s informal claims onto the formal claims Scyther checks. Every claim verifies, at both bounded (5 runs) and unbounded search, on the protocol as built above. Run against the prefix sketch instead – the version missing the responder’s identity from the initiator’s signature – the identical tool and claims find a real attack: a non-injective agreement violation in which the two parties finish believing they talked to different counterparts. That the tool is silent on the fixed protocol and vocal on the broken one is a stronger confirmation than a clean run alone would be. Two limitations are stated rather than elided: the seen-nonce cache is stateful and has no representation in Scyther’s per-session symbolic model, so replay resistance rests on the unit tests, not this proof; and Scyther has no equational theory for $g^{ab} = g^{ba}$, so the Diffie-Hellman exchange is modelled with the standard oracle idiom Scyther’s own author uses for the same purpose, rather than checked natively. Full model, tool provenance, and raw output: verification/.

C. The honeypot feature pipeline: defining Sig_{RW} and FT_{RW}

Algorithm 2 states, in its entirety, that a signature Sig_{RW} and a feature vector FT_{RW} are “generated” (lines 5–6) from data a ransomware honeypot collects. Nothing else is said: not what the honeypot observes, not what a signature contains, not what a single feature is. This gap is consequential rather than cosmetic. Without a defined FT_{RW} , Algorithm 3’s detection module has nothing to train on, and §VII’s evaluation silently substitutes an unrelated dataset instead (§V-A). Supplying this layer is the largest design contribution of this project: it is the one piece of the paper’s own specification that, once built, lets Phase 3 actually consume Phase 2’s output, which the paper as published cannot do.

Two distinct notions hide under one symbol. We split Sig_{RW} into a content_digest (a SHA-256 hash over the sample’s canonical behavioural trace – the malware-

TABLE I
§V-1’s informal claims, the formal claim Scyther checks for each, and the result.

§V-1 claim	Formal claim	Result
Adversary cannot estimate SK	Secret	Verified
Mutual auth., $A \rightarrow B$	Niagree/ Nisynch	Verified
Mutual auth., $B \rightarrow A$	Niagree/ Nisynch	Verified
Distinct keys per session	fresh-nonce gument	Holds by constr.
Impersonation resistance	Weakagree/ Alive	Verified
No reflection	default self-play search	No attack found
Cross-id. (fixed)	replay Niagree, fixed model	Verified
Cross-id. (pre-fix)	replay Niagree, pre-fix model	Attack found
Within-window replay	– (stateful cache)	Out of scope

signature sense, used for identification) and an attestation (an ECDSA signature by CS_i over that digest, a timestamp, and a collector id – the digital-signature sense, used for on-chain authenticity). The paper’s line 5 conflates a content identifier with a cryptographic signature; separating them is what makes “building a signature for a malicious program” mean something coherent rather than a category error.

FT_{RW} is a 22-feature, seven-group behavioural vector – filesystem, entropy, crypto-API, process, network, persistence, and kill-chain-stage groups – grounded in the measurement categories the paper’s own comparison detectors report. Two decisions inside it are load-bearing rather than incidental. First, the kill-chain-stage group is truncated to an observable four-stage prefix (arrival, enumeration, bulk transform, cleanup). The paper’s own seven-stage kill chain ends notification $\rightarrow$ payment $\rightarrow$ decryption, stages that only ransomware ever reaches; an untruncated “stage reached” feature would be the label wearing a feature’s clothes, trivially found by every model, and worth nothing as a measure of detection. Second, because we wrote both the sample generator and the classifier that consumes it, an easy corpus would report a number that describes the generator, not any detector. We therefore built the corpus to a stated difficulty. A quarter of every draw (AMBIGUOUS_FRACTION = 0.25) comes from confusable pairs – two profiles sharing one identical parameter set – so those samples carry no label information at all, putting a floor of 0.125 under the Bayes error before any other class overlap is counted. Four of the benign profiles (a backup agent, a disk-encryption tool, an installer, a cleanup utility) are deliberately ransomware-shaped, and every counter is noised, with whole sensor groups going unobserved at a rate independent of the label. The corpus therefore carries a stated intended ceiling, EXPECTED_BAYES_ACCURACY $\approx$ 0.85, against

TABLE II

FT_{RW} : seven feature groups, 22 features total. The kill-chain group is truncated to the observable prefix; the full walk is retained in provenance data, outside the vector.

Group	Representative features
Filesystem	files touched/s, read:write ratio, rename rate, extension-change rate, traversal breadth
Entropy	mean/variance of written-block entropy, entropy delta pre/post write
Crypto API	crypto call count, key-generation events, API call n -grams
Process	child spawns, injection attempts, privilege-escalation attempts
Network	C2 beacon count, DNS entropy, outbound burst rate
Persistence	registry/autostart writes, shadow-copy deletions, backup-path access
Kill-chain (trunc.)	observed stages (capped at 4), mean stage dwell time

which any later detector result must be read: far above it is a leak, not a success. Table II lists the seven groups and representative members.

The leakage tests this difficulty budget requires caught a real problem before the corpus shipped. An early parameter set put a single feature, extension-change rate, at 0.885 separability on its own – an unintended single-feature giveaway, exactly what a test requiring every feature and group mean to stay under 0.90 separability exists to catch. After retuning, the most separable single feature on the committed corpus is extension-change rate at 0.816, followed by rename rate (0.803) and observed-stage count (0.796); a simple Gaussian baseline trained on the training draw and scored on the held-out evaluation draw reaches 0.830 balanced accuracy – comfortably under the intended 0.85 ceiling, which is the point, not a shortfall.

D. Preprocessing: what “remove the abnormalities” can-not mean

Algorithm 2 line 3 pre-processes honeypot data and, without qualification, “removes the abnormalities.” Read literally as statistical outlier removal, the instruction is self-defeating on exactly this corpus: ransomware behaviour is the minority class by construction – it is the abnormality – so an outlier-removal step deletes the samples the detector exists to learn from. Applied naively, Algorithm 2’s own pre-processing would hand Algorithm 3 a clean-looking dataset with the positive class silently gone, and §VII would then report a plausible accuracy computed on data that never saw an attack. §V-D develops this as a critique; here we record the design decision it forces. We read “abnormalities” as sensor defects, never unusual behaviour, and implement five explicit rules: drop structurally empty records (no observed stage, or a non-positive duration); clamp impossible values to their physical range, but mark a negative count unreadable rather than clamp it, because “the sensor returned nonsense” and “the program did nothing” are different facts that

must not collapse into the same number; deduplicate byte-identical traces; mark missing counters as missing rather than impute them, so the honeypot’s blind spots stay visible to later stages instead of being hidden inside invented data; and normalise absolute counts into per-second rates. No outlier filtering, smoothing, or class re-balancing is performed. A dedicated test pins the property that matters: cleaning never removes the positive class.

E. Per-system backup index

Algorithm 5, lines 1–2, imply a linear scan of BC_{DTBU} to find one system’s backups – $O(\text{chain length})$. We add a per-system index, maintained by the key-holding cloud server on every block append (system_id is itself encrypted, so only that server can build it), reducing lookup to $O(1)$ while preserving the paper’s semantics exactly: a stale or cold index falls back to a full scan, and the index stores only pointers, never plaintext, so restore still re-reads and re-verifies every block it points to. This moves the decryption cost from recovery time to append time; it does not remove that cost. Index maintenance is kept explicitly outside Algorithm 1’s timed path, since the paper’s own Algorithm 1 has no such index and folding ours into the benchmark would silently advantage our own Fig. 6(a) numbers against the paper’s.

F. Case-3: declining automated ransom payment

Algorithm 4 line 7 specifies that when the demanded ransom is less than the assessed value of a system’s data, the framework pays it and retrieves the decryption key – fully automated. We decline to implement the payment path. The paper’s own §II-C admits there is no guarantee the key arrives or works even if payment is made, and automating ransom payment to an active attacker is sanctioned conduct in several jurisdictions independent of whether it works. What we build instead preserves the control-flow contribution without the harmful capability: the branch evaluates the paper’s own condition, emits a PolicyDecision object, writes exactly one audit log record, and terminates in a simulated POLICY_BLOCKED state regardless of the decision’s outcome. No network access, wallet interaction, or transaction construction exists anywhere in the module that implements this, enforced by a test that statically scans it for exactly those imports and asserts the function’s only observable effect is the one log record.

IV. Reproduction Results

Every number in this section is directly measured, with a recorded configuration hash, seed, and run log behind it. We report paper-mode and honest-mode evaluations side by side throughout; neither is presented without the other.

A. Paper-mode reproduction (Targets 1–2)

Table III reproduces Table II exactly as published, with two additions: the source dataset for each comparison technique (rows 1–4 are evaluated on network traffic traces, dynamic-analysis logs, an unpublished corpus, and PE-file features respectively – none of them BitcoinHeist, so the comparison in the paper’s own table is not like-for-like), and our own measured BSFR-SH row.

Our reproduction follows the paper’s own protocol exactly: BitcoinHeist resampled to 90% ransomware / 10% benign – bounded, it turns out, at $n = 46,014$ rows, 1.6% of the 2,916,697-row dataset the paper cites two paragraphs earlier, a fact the paper never states and that the resample’s own arithmetic proves without touching the data – address dropped as an identifier, and an address-grouped train/test split. Grouping is the methodologically correct choice once we confirmed, directly against the raw file, that every address carries exactly one label end to end and that ransomware addresses repeat at roughly twice the rate benign ones do; a random split lets near-duplicate rows leak across the train/test boundary at a rate the 90% skew amplifies. Under this protocol our best model, random forest, reaches 0.9442 accuracy / 0.9697 F1 – 4.56 accuracy points and 2.93 F1 points below the published 0.9898/0.990. The paper’s own best-reported algorithm, decision tree, reaches only 0.9224/0.9569 in our hands.

The baseline every number in this table must be read against: on the paper’s own 90/10 split, a classifier that answers “ransomware” unconditionally, looking at no feature at all, scores 0.9000 accuracy and 0.9474 F1. Sharmeen et al.’s published 0.9596/0.960 – presented in Table II as a competitive prior technique – clears that do-nothing baseline by only 0.013 F1.

Before accepting 4.56 points as evidence of a genuine detector deficit rather than an artifact of our own pipeline, we tested the two most plausible leakage sources: address kept as a naively label-encoded feature, and a non-grouped (random) train/test split. Table IV reports the resulting 2×2 ablation for random forest, the best model throughout. Both sources are real – keeping address under a random split moves decision tree by +0.92 points over its own grouped-split baseline, and that effect vanishes under grouping, the expected signature of an identifier memorised across a train/test boundary – and neither is large enough to matter alone. Stacking both does not close the gap: the best of all sixteen model $\times$ cell combinations tested reaches 0.9540/0.9749, still 3.58 accuracy points and 2.41 F1 points under the published figure. We report the residual gap as unexplained with a measured bound: we tested the two candidates available to us, ruled out neither as fully responsible, and did not invent a third.

B. Honest-mode evaluation (Targets 1–2)

Table V reports the full-scale (2,916,697-row) honest_mode evaluation: natural class balance, all four models run to completion, precision, recall, PR-AUC, MCC

and minority-class F1 rather than accuracy – because accuracy is exactly the metric the paper’s 90/10 resample was constructed to inflate.

The constant-negative baseline – never predict ransomware – scores 0.9858 accuracy at the dataset’s natural class balance. We report this strictly as an illustration of how distribution-sensitive accuracy is, not as a head-to-head against the paper’s published 98.98%: the two numbers are computed under different class distributions, and a constant classifier’s accuracy is a direct function of the positive rate it is scored against. The commensurable comparison is Table III’s within-split baseline above.

Logistic regression makes the honest-mode point in one model. Under the paper’s 90/10 resample it predicts everything positive (0.9002/0.9475, indistinguishable from the constant-positive baseline). At the dataset’s natural balance, same declared hyperparameters, same seed, it predicts nothing positive (precision, recall, and MCC all 0). Class balance alone flips the same model from all-positive to all-negative; neither behaviour reflects a detector that has learned anything about ransomware addresses.

Random forest is the strongest of the four at natural balance (MCC 0.457, recall 0.286 – it recovers roughly 29% of ransomware addresses at 74% precision), followed by decision tree (MCC 0.401) and KNN (MCC 0.365). KNN’s full-dataset run, submitted to the university’s Ada HPC cluster because its memory footprint at 2.9M rows exceeds our 8 GB development machine, scored meaningfully higher (MCC 0.365) than an earlier, memory-constrained 780,336-row local subsample (MCC 0.300) – more data helps KNN too, a comparison the paper’s own 46K-row resample never had the chance to make either way. The HPC job itself is a useful data point about the paper’s own setup: a single-model, full-scale KNN fit at natural class balance ran 36 minutes on a general-purpose research cluster allocation, against a paper that reports its entire evaluation – four models, unspecified cross-validation – as having run on an 8 GB laptop. Either the paper’s own KNN never saw more than a small fraction of the 2.9M rows it describes, consistent with the 90/10 resample’s 46,014-row bound, or its reported runtime is missing from §VII entirely; §VII gives no timing figures for the ML evaluation at all, only for the separate consensus benchmark.

C. Blockchain timing (Targets 3–4)

Table VI reports Targets 3–4 at the paper’s own three cases (5/10/15 blocks $\times$ 100 transactions). Absolute seconds are two to three orders of magnitude below the paper’s – expected, given Python on commodity hardware against the paper’s Java-on-Windows measurement, and not a claim we contest. The comparison we make is trend and ratio, not seconds.

The paper’s headline shape – time growing sub-linearly, TPS rising as the chain grows – is not reproduced. Figure 1

TABLE III

Table II reproduction: published comparison techniques, published BSFR-SH row, and our measured BSFR-SH row under the paper’s own 90/10 protocol. $n = 46,014$ for every BitcoinHeist row – the resample’s own size bound, not a design choice.

Technique	Mode	Dataset	Accuracy	F1	n
Almashhadani et al. [4]	paper_reported	network traffic traces (Locky)	0.9708	0.971	–
Hwang et al. [5]	paper_reported	dynamic analysis logs	0.9730	0.973	–
Sharmeen et al. [6]	paper_reported	semi-supervised, own corpus	0.9596	0.960	–
Bae et al. [7]	paper_reported	PE-file features	0.9865	0.987	–
BSFR-SH (paper, best of 4: decision tree)	paper_reported	BitcoinHeist addresses	0.9898	0.990	46,014
BSFR-SH (ours, random forest)	measured	BitcoinHeist addresses	0.9442	0.9697	46,014
BSFR-SH (ours, decision tree)	measured	BitcoinHeist addresses	0.9224	0.9569	46,014
BSFR-SH (ours, logistic regression)	measured	BitcoinHeist addresses	0.9002	0.9475	46,014
BSFR-SH (ours, k -NN)	measured	BitcoinHeist addresses	0.8892	0.9410	46,014
Constant-positive baseline	computed	BitcoinHeist addresses (90/10 split)	0.9000	0.9474	46,014

TABLE IV

Leakage ablation (Q10), random forest, address dropped/kept $\times$ grouped/random split, all cells at identical declared hyperparameters and seed.

Cell	Acc.	F1	vs. published
address dropped $\times$ grouped	0.9442	0.9697	–4.56 <i>pt</i>
address dropped $\times$ random	0.9479	0.9717	–4.19 <i>pt</i>
address kept $\times$ grouped	0.9484	0.9719	–4.14 <i>pt</i>
address kept $\times$ random	0.9540	0.9749	–3.58 pt

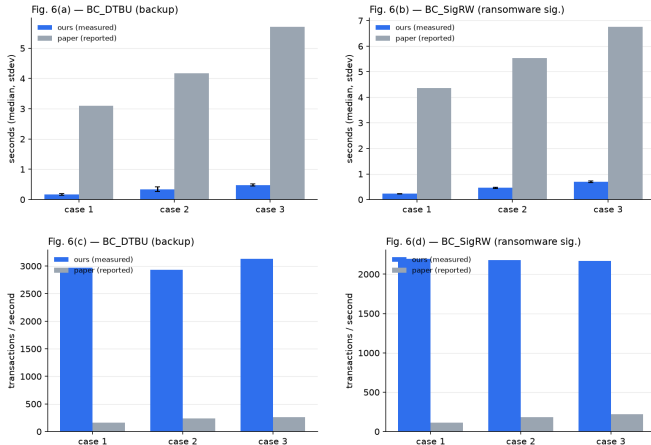


Fig. 1. Figs. 6(a)–(d) reproductions: measured time and derived TPS, both chains, against the paper’s published values. Our time grows near-linearly with block count rather than sub-linearly, and our TPS is flat rather than rising – the direct consequence of the flat marginal cost in Table VII.

shows measured time growing essentially linearly with block count on both chains, and Table VII makes the reason explicit: our harness starts each block’s timer only after the cluster is already constructed, so no fixed setup cost sits inside the measured span to amortise. Per-block marginal cost is flat – within measurement noise, the same at case 1 (5 blocks committed) as at case 3 (15 blocks committed) – on both chains, and TPS is correspondingly flat rather than rising. This is a second, independent line of evidence for the claim that the paper’s own published TPS values equal tx/time exactly, verified against all six of its own data points: the first line of evidence is that arithmetic

identity; this one is a measurement of what the same arithmetic looks like once the fixed cost is structurally excluded from the timed span, and it produces exactly the flat curve the amortisation account predicts.

BC_{SigRW} is consistently slower than BC_{DTBU} , by 35–44% across the three cases (paper: 18–41%). We attribute this structurally, not by tuning to match, to one additional ECDSA signature per transaction – the honeypot pipeline’s attestation (§III-C) – plus a many-small-floats payload encoding that costs more per byte to canonicalise than the backup chain’s single opaque byte blob.

V. Critique

Five defects, ordered by severity rather than by the order we found them. They are the ones that rise to a methodological or architectural problem; a complete record of every smaller departure from the paper’s published design, each with its own rationale, is maintained separately and is not reproduced here.

A. FLAW-2 – framework/evaluation disconnect

Algorithm 2 collects data from a ransomware honeypot and Algorithm 3 trains a detector on the resulting signatures and behavioural features – properties of an executing, or emulated, program. Section VII then evaluates entirely on BitcoinHeist: address-level graph features (length, weight, count, looped, neighbours, income) of a Bitcoin address that has already received a ransom payment. These are different problems. Detecting a running ransomware program from its behaviour and forensically labelling a wallet address after the fact share no features, no data-generating process, and arguably not even the same notion of a positive instance – one is prospective and behavioural, the other retrospective and financial. Nothing in §IV-C or §VII acknowledges the substitution.

We built both pipelines rather than choosing one. The BitcoinHeist pipeline (§IV-A–§IV-B) reproduces the paper’s actual published evaluation. The honeypot pipeline (§III-C, §VI) runs Algorithm 2’s honeypot output through real Phase-2 consensus onto BC_{SigRW} , decrypts it back out, and trains Algorithm 3’s four models on it – the sequence the paper’s own Fig. 3 draws. It produces a real, bounded result (0.8422 balanced accuracy, §VI). The

TABLE V

Honest-mode evaluation: natural 1.42% ransomware prevalence, full 2,916,697-row dataset, all four models (KNN run on Ada HPC; job 2700090). Accuracy is intentionally not reported here – see text.

Model	n	Precision	Recall	PR-AUC	MCC	F1 (minority)
Random forest	2,916,697	0.7442	0.2864	0.4646	0.4573	0.4136
Decision tree	2,916,697	0.3972	0.4231	0.1797	0.4012	0.4097
k -NN	2,916,697	0.6008	0.2286	0.2242	0.3654	0.3312
Logistic regression	2,916,697	0.0000	0.0000	0.0177	0.0000	0.0000
Constant-negative baseline	2,916,697	undef. [†]	0.0000	0.0142	0.0000	0.0000

[†]Precision is 0/0: a constant-negative classifier makes no positive predictions, so there is no denominator – undefined, not zero.

Constant-negative accuracy (illustration only, not head-to-head with the paper’s 98.98%): 0.9858.

TABLE VI

Target 3/4: measured vs. published time and derived TPS, both chains, all three cases.

Case / chain	Time (ours)	Time (paper)	TPS (ours)
Case 1, BC_{DTBU}	0.168 s	3.10 s	2972
Case 1, BC_{SigRW}	0.228 s	4.36 s	2195
Case 2, BC_{DTBU}	0.341 s	4.17 s	2931
Case 2, BC_{SigRW}	0.459 s	5.54 s	2179
Case 3, BC_{DTBU}	0.479 s	5.71 s	3132
Case 3, BC_{SigRW}	0.691 s	6.76 s	2169

TABLE VII

Mean measured per-block marginal cost by case (three samples per cell, seconds), the direct evidence for DEV-08. If the paper’s amortisation account were absent from our own harness by construction, marginal cost would fall from case 1 to case 3; instead it is flat.

Chain	Case 1	Case 2	Case 3
BC_{DTBU}	0.0335 s	0.0334 s	0.0316 s
BC_{SigRW}	0.0455 s	0.0450 s	0.0450 s

paper’s framework half, run end to end as specified, was never demonstrated by its authors; only the mismatched evaluation half was. FLAW-2 is therefore sharper than “the paper evaluates on the wrong data”: the paper’s own framework, when actually built and run, produces a defensible result that the paper itself never reports, choosing an unrelated dataset instead.

B. FLAW-4 – evaluation methodology

Section VII resamples BitcoinHeist to 90% ransomware before evaluating, without stating that the resample is bounded by the scarce class: only 41,413 ransomware rows exist in the cited 2,916,697-row dataset, so a 90%-ransomware sample can hold at most $41,413/0.9 = 46,014$ rows – 1.6% of the dataset described two paragraphs earlier. The published 98.98% accuracy is a number from a $\sim 46K$ -row experiment, reported with nothing to mark that it is not the 2.9M-row one. Compounding this, the resulting class balance makes accuracy nearly free: a constant-positive classifier on this split scores 90.0% (§IV-A).

Reproducing the paper’s own protocol exactly, our best model reaches 0.9442/0.9697, 4.56 accuracy points under published. An ablation over the two candidate leakage sources we could identify – address retained as a feature,

and a non-grouped split – narrows this only to 3.58 points at the single most leakage-favourable cell tested (Table IV), and both effects vanish or shrink under the methodologically correct grouped split. We report the residual gap as unexplained, with a measured bound, rather than attribute it to a cause we found no evidence for.

C. FLAW-5 – the §V-3 security claim

Section V-3 argues that BSFR-SH resists 51% attacks because it replaces proof-of-work with pBFT. The argument borrows PoW’s 51% threshold and applies it to a consensus mechanism with a different one: pBFT is safe only while fewer than $n/3$ of replicas are Byzantine ($n \geq 3f + 1$). Moving from PoW to pBFT does not raise the fraction of the network an adversary must control – it lowers it, from just over one half to one third. At the paper’s own deployment of four miner nodes, $f = 1$: two colluding replicas (50% of the cluster – above the $f = 1$ bound the cluster actually tolerates, and below the 51% figure the paper cites as the threshold being defended against) can make two honest replicas commit different blocks at the same height. We demonstrate this directly rather than argue it: a test constructs exactly this scenario against our own pBFT implementation and asserts the fork occurs, at exactly the bound f predicts and no tighter.

The fair counter-argument, which §V-3 could have made and does not, is that pBFT’s threshold counts identities, not hash power, and in a permissioned deployment identities cannot be minted by an outside adversary – a property we test separately, showing a swarm of unauthorised identities cannot form a certificate regardless of size. The defensible version of §V-3’s claim is therefore operational – “an attacker must compromise two of our four named cloud servers” – not consensus-theoretic, and it gets weaker, not stronger, in relative terms as a reason to prefer pBFT over PoW at a fixed and small node count.

D. DEV-26 – preprocessing deletes the positive class

Algorithm 2 line 3 pre-processes honeypot data and, in its entirety, “removes the abnormalities.” The literal reading is statistical outlier removal. On a corpus whose purpose is teaching a detector what ransomware looks like, that reading is self-defeating: ransomware behaviour is the

TABLE VIII

On-chain storage growth, daily 1 MiB backups, 4 replicas. Every figure is arithmetic over §IV-C’s measured overhead factor, not a new measurement.

Scenario	Per-node/yr	Cluster/yr ($\times 4$)	Time to 1 TiB/node
Small clinic (50 pts)	20.3 GiB	81.3 GiB	≈ 50.4 yr
Medium hospital (5,000 pts)	1.98 TiB	7.94 TiB	≈ 6.09 mo
Large network (50,000 pts)	19.8 TiB	79.4 TiB	≈ 8.4 days

minority class by construction – it is the abnormality – so an outlier-removal step deletes exactly the samples the detector exists to learn from. Applied naively, Algorithm 2’s own pre-processing step would hand Algorithm 3 a clean-looking dataset with the positive class silently gone, and §VII would then report a plausible-looking accuracy computed on data that never saw an attack. We interpret “abnormalities” as sensor defects rather than unusual behaviour (§III-D) and pin the alternative reading’s failure mode with a direct test asserting that cleaning never removes the positive class.

E. DEV-08 – TPS is amortisation, not throughput

Section VII-D presents TPS as a measured throughput property that improves as the chain grows across the paper’s three cases. Every one of the six published TPS values equals $\text{total_tx}/\text{total_time}$ exactly, verified to four significant figures against all six points; the rise the paper reports is the amortisation of a fixed per-run setup cost over a growing number of blocks, not a throughput property of the consensus mechanism. We add a second, measured line of evidence beyond this arithmetic observation: when we time only the append span itself, excluding cluster setup, our own marginal per-block cost is flat across cases 1 through 3 on both chains (Table VII), and our own TPS is correspondingly flat rather than rising (§IV-C). Removing the fixed cost from the timed span removes the rise – exactly what “the rise is amortisation” predicts, and what our own implementation lets us test directly rather than infer.

F. GAP-2 – storage-cost analysis

Section VII benchmarks consensus timing but never asks how much storage its own design consumes, or at what scale that stops being feasible. We compute this from values already measured elsewhere in this project – the declared 4096B transaction payload (DEV-15), the measured 215B framing and 161B hybrid-encryption overhead per chunk (DEV-24), and the measured $1.14\times$ full-block overhead factor (header, Merkle root and canonical encoding, amortised over a 100-tx block, §IV-C) – rather than from new estimates; full arithmetic in docs/STORAGE_ANALYSIS.md.

At hospital scale, the paper’s own on-chain design fills 1 TiB per replica in about six months under daily backups; at network scale, in under three weeks. Backup frequency is the larger lever of the two the paper leaves open: at

the medium scenario, daily backups cost $\approx 7\times$ weekly and $\approx 30\times$ monthly in chain growth for identical clinical content, a bigger effect than DEV-15’s payload-size sweep found from a $16\times$ range of payload sizes alone.

Against this, an on-chain-hash / off-chain-data design – store only a content digest and a storage pointer on-chain, keep the encrypted payload in external storage – preserves the same tamper-evidence the paper’s Merkle-root-and-signature chain already provides (recomputing and comparing the digest catches off-chain tampering exactly as the paper’s own immutability argument requires), at ≈ 504 bytes per backup regardless of payload size, against ≈ 1.14 MiB for a 1 MiB backup under the paper’s design – a $\approx 2,371\times$ reduction that widens further as backups get larger, since the hash-only cost is flat and the paper’s is not. What it does not preserve is on-chain availability: lose the off-chain store and the digest alone cannot reconstruct the backup, where the paper’s design always can. We state this trade-off rather than resolving it, per this project’s rule against silently revising the paper’s design.

Compute is not the bottleneck this analysis surfaces: the large-network scenario’s daily block count, at §IV-C’s measured marginal per-block cost, comes to $\approx 4,045$ CPU-seconds/day ($\approx 4.7\%$ of a day) on the same 8 GB dev laptop – comfortably inside its measured throughput ceiling, though nothing here measures a production cluster’s network or disk behaviour, only this one machine’s compute.

G. What the paper gets right

Two of the paper’s choices are execution defects, not premise defects, and the distinction matters for how it should be read. The two-chain architecture – one chain for the ground truth a detector trains on, a separate chain for the backups a system restores from, never sharing node sets or state – genuinely does separate detection integrity from recovery integrity: an adversary who compromises the mechanism protecting one does not thereby compromise the other. And the central idea that blockchain immutability can protect a detector’s training data from tampering, in a domain where an attacker has every incentive to poison exactly that data if they can reach it, is sound, and is not the load-bearing idea of any of the four papers Table II compares against. The paper’s execution of that idea – its evaluation methodology, its security argument for choosing pBFT, its preprocessing specification, and the connection between what it builds and what it measures – has the five gaps above. Both are true, and a critique that lists only the five would misrepresent the paper’s contribution as smaller than it is.

VI. Honeytrap Detection Evaluation

§III-C described the design of the honeytrap feature pipeline the paper omits. This section reports what it measures once built end to end: honeytrap deployment $\rightarrow$

TABLE IX

Honeypot detection result (Alg. 3 end to end) and its derived confusion matrix. TP/FP are computed from the reported precision, recall and evaluation-set composition (353 of 731 eval records positive) – a computed quantity, not a separately measured one.

Train n / Eval n	1467 / 731
Balanced accuracy	0.8422
Precision	0.8439
Recall	0.8272
MCC	0.6849
PR-AUC	0.9100
Constant-positive baseline	0.5000
Derived confusion matrix, eval set ($n = 731$)	
True positive	292
False negative	61
False positive	54
True negative	324

Phase-2 collection $\rightarrow$ real four-node pBFT consensus onto $BC_{SigRW} \rightarrow$ Phase-3 decryption, training, and detection – the sequence Fig. 3 of the paper draws and the paper’s own evaluation never runs.

Table IX reports the result on the committed corpus (1467 training records, 731 evaluation records, drawn from two independent seeds so no latent generator parameter is shared across the split): 0.8422 balanced accuracy, between the measured simple-Gaussian baseline (0.830) and the corpus’s stated intended ceiling (≈ 0.85 , §III-C) – above the naive baseline, not above the ceiling that would indicate the classifier is reading a leak rather than the corpus’s designed difficulty. The evaluation script checks this automatically against a 0.05 leak margin and refuses to record a result that would fail the check; it did not fire here. The derived confusion matrix shows a detector that is neither trivially accurate nor degenerate: 292 of 353 ransomware records correctly flagged, 61 missed, 54 benign records misclassified.

This is the first number in this project produced by Phase 2’s output feeding Phase 3’s input through the paper’s own architecture. It demonstrates that the framework, as specified, is not merely completable on paper – once §IV-B’s undefined $SigRW/FT_{RW}$ are actually given a definition, Algorithms 2 and 3 connect into a working, non-degenerate detection pipeline. This is the strongest available rebuttal to a reading of FLAW-2 as “the framework as designed cannot work”: it can. The paper simply never tried.

VII. Conclusion

We built a complete, from-scratch implementation of BSFR-SH’s five algorithms, both of its independent private blockchains under real pBFT consensus, and both a faithful reproduction of the paper’s published evaluation and the honeypot feature pipeline its own Phase 2/3 requires but never defines. Reproducing the paper’s own methodology exactly, its headline detection result does not reproduce: our best model reaches 0.9442/0.9697 against a published 0.9898/0.990, and an ablation over

both candidate leakage sources we identified narrows but does not close a 3.58-point residual gap. We find five further, independent defects – a disconnect between what the framework is designed to detect and what it is evaluated on, a degenerate and undisclosed evaluation split, a security argument that inverts pBFT’s actual threshold relative to proof-of-work, a preprocessing specification that, read literally, deletes the class it exists to detect, and a reported throughput curve that is arithmetic amortisation rather than a measured property. Against this, the paper’s central architectural idea – two chains, separately protecting a detector’s ground truth and a system’s recoverability, from an adversary with every incentive to attack both – is sound, and our own honeypot pipeline shows the framework can produce a real, bounded detection result (0.8422 balanced accuracy) once its undefined layer is actually built.

What remains open: the 3.58-point gap between our best reproduction and the published figure has a measured lower bound but no identified cause – neither address-as-feature nor split strategy, alone or together, explains it, and no further candidate was tested for lack of one. We did not pursue the stretch extensions in the project roadmap – a formal (Scyther) model of the session-establishment protocol, adversarial feature-space evasion against the detector, a storage-cost analysis for on-chain healthcare backups at realistic scale, an asynchronous pBFT variant with modelled network latency, and hybrid-blockchain support, all of which the paper’s own conclusion or our design work flags as open – since they fall outside this report’s scope.

Taken together, the two findings point in the same direction rather than opposite ones. A paper whose evaluation methodology cannot be checked because its protocol is underspecified, and whose one checkable headline number does not reproduce even under the most literal reading of its own stated split, is a paper whose central empirical claim should not be relied on as published. That is a narrower conclusion than “the framework does not work”: our own honeypot pipeline shows a version of Algorithm 3, trained on data Algorithm 2 actually specifies producing, detecting ransomware at a real, bounded, non-degenerate rate. The architecture is worth building on. The published numbers, as published, are not evidence that it works as well as claimed.

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